

- # Characteristics of strategic information

INTEGRATED	Must have a single, enterprise-wide view.
DATA INTEGRITY	Information must be accurate and must conform to business rules.
ACCESSIBLE	Easily accessible with intuitive access paths, and responsive for analysis.
CREDIBLE	Every business factor must have one and only one value.
TIMELY	Information must be available within the stipulated time frame.

Figure 1-2 Characteristics of strategic information.

Operational Systems

Get the data in

Making the wheels of business turn

- ◆ Take an order
- ◆ Process a claim
- ◆ Make a shipment
- ◆ Generate an invoice
- ◆ Receive cash
- ◆ Reserve an airline seat

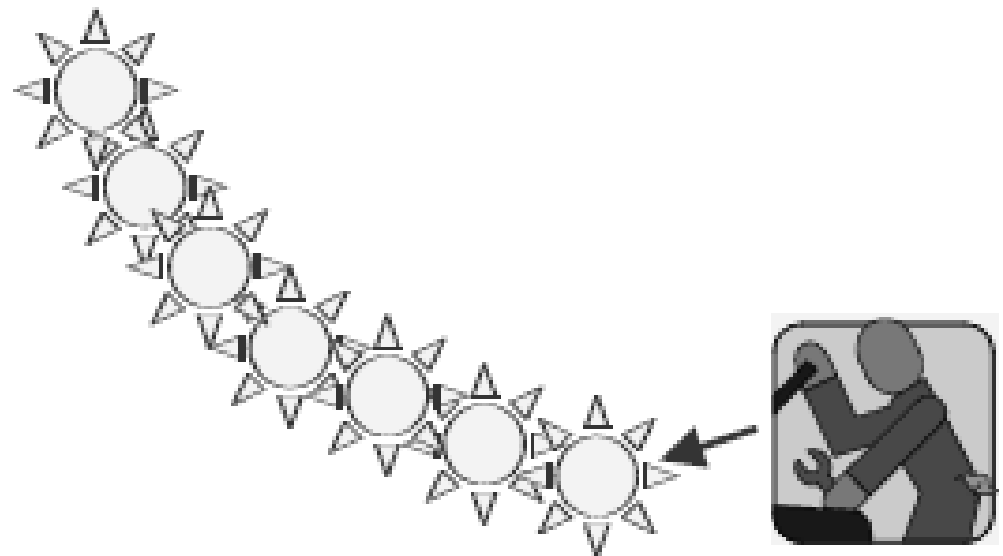


Figure 1-5 Operational systems.

Decision Support systems

Get the information out

Watching the wheels of business turn

- ◆ Show me the top-selling products
- ◆ Show me the problem regions
- ◆ Tell me why (drill down)
- ◆ Let me see other data (drill across)
- ◆ Show the highest margins
- ◆ Alert me when a district sells below target

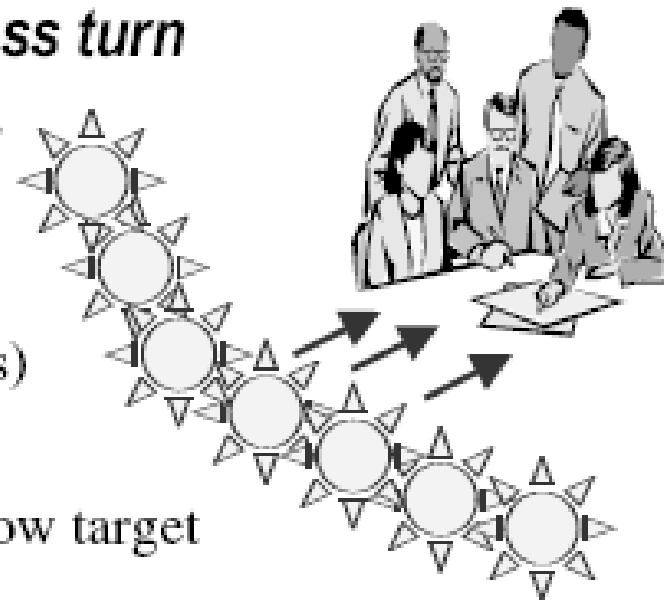


Figure 1-6 Decision-support systems.

- These operational systems (order processing, inventory control, claims processing, outpatient billing, ...) are not designed or intended to provide strategic information.
- If we need the ability to provide strategic information, we must get the information from altogether different types of systems.
- Only specifically designed decision support systems or informational systems can provide strategic information.

How are they different?

	OPERATIONAL	INFORMATIONAL
Data Content	Current values	Archived, derived, summarized
Data Structure	Optimized for transactions	Optimized for complex queries
Access Frequency	High	Medium to low
Access Type	Read, update, delete	Read
Usage	Predictable, repetitive	Ad hoc, random, heuristic
Response Time	Sub-seconds	Several seconds to minutes
Users	Large number	Relatively small number

Figure 1-7 Operational and informational systems.

• **Processing Requirements in the New Environment**

- Most of the processing in the new environment for strategic information will have to be analytical.

There are four levels of analytical processing requirements:

1. Running of simple queries and reports against current and historical data
2. Ability to perform “what if ” analysis in many different ways
3. Ability to query, step back, analyze, and then continue the process to any desired length
4. Spot historical trends and apply them for future results

- Data warehousing concept

- Take all the data you already have in the organization, clean and transform it, and then provide useful strategic information.

•Data warehousing concept

- One of the most important approaches to the integration of data sources is based on a *data warehouse* architecture.
- In this architecture, data coming from multiple external data sources (EDSs) are extracted, filtered, merged, and stored in a central repository, called a data warehouse (DW).
- Data are also enriched by historical and summary information. From a technological point of view, a data warehouse is a huge database from several hundred GB to several dozens of TB.

• **Functional definition of the data warehouse**

- The data warehouse is an informational environment that:
 - Provides an integrated and total view of the enterprise
 - Makes the enterprise's current and historical information easily available for decision making
 - Makes decision-support transactions possible without hindering operational systems
 - Renders the organization's information consistent
 - Presents a flexible and interactive source of strategic information

DWH – Blend of many technologies

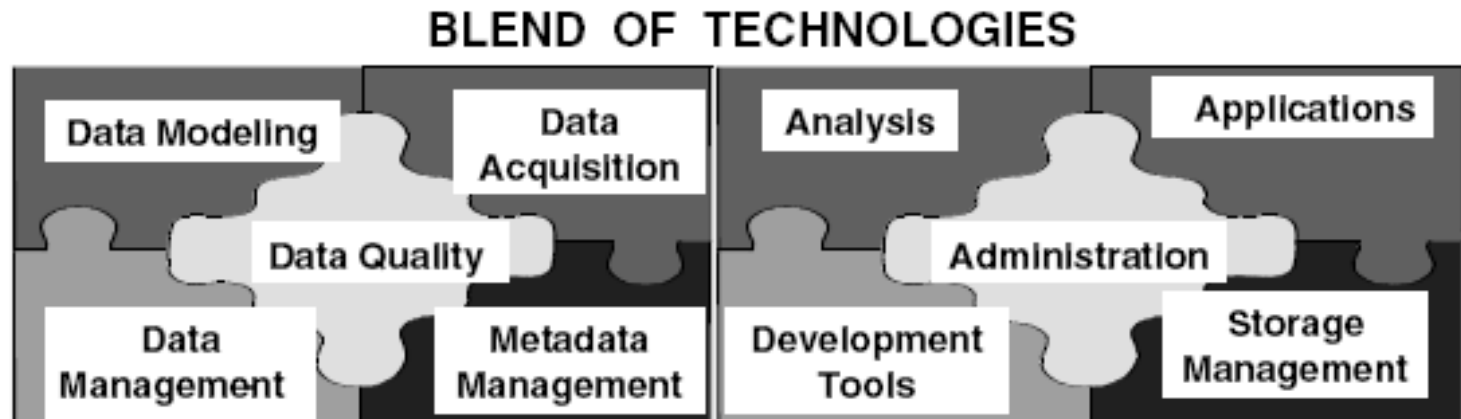
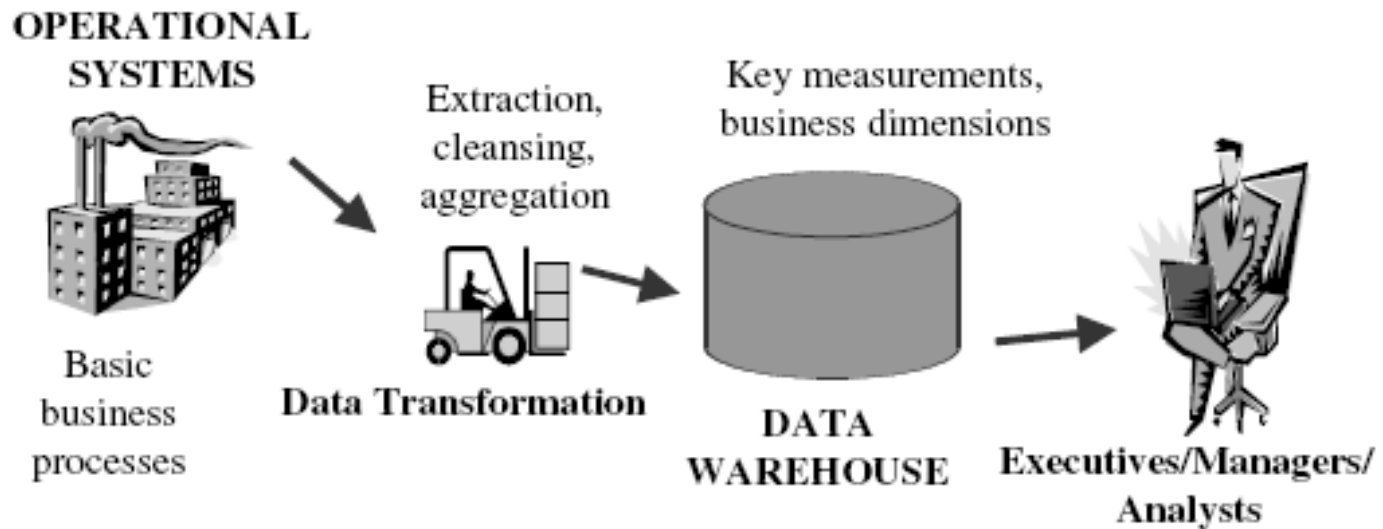


Figure 1-9 The data warehouse: a blend of technologies.

- Bill Inmon's definition

- Bill Inmon, considered to be the father of Data Warehousing provides the following definition:

- “A Data Warehouse is a **subject oriented, integrated, nonvolatile, and time variant** collection of data in support of management's decisions.”

- The data in the data warehouse is

- Separate
- Available
- Integrated
- Time stamped
- Subject oriented
- Nonvolatile
- Accessible

•Subject-oriented

In the data warehouse, data is not stored by operational applications, but by business subjects.

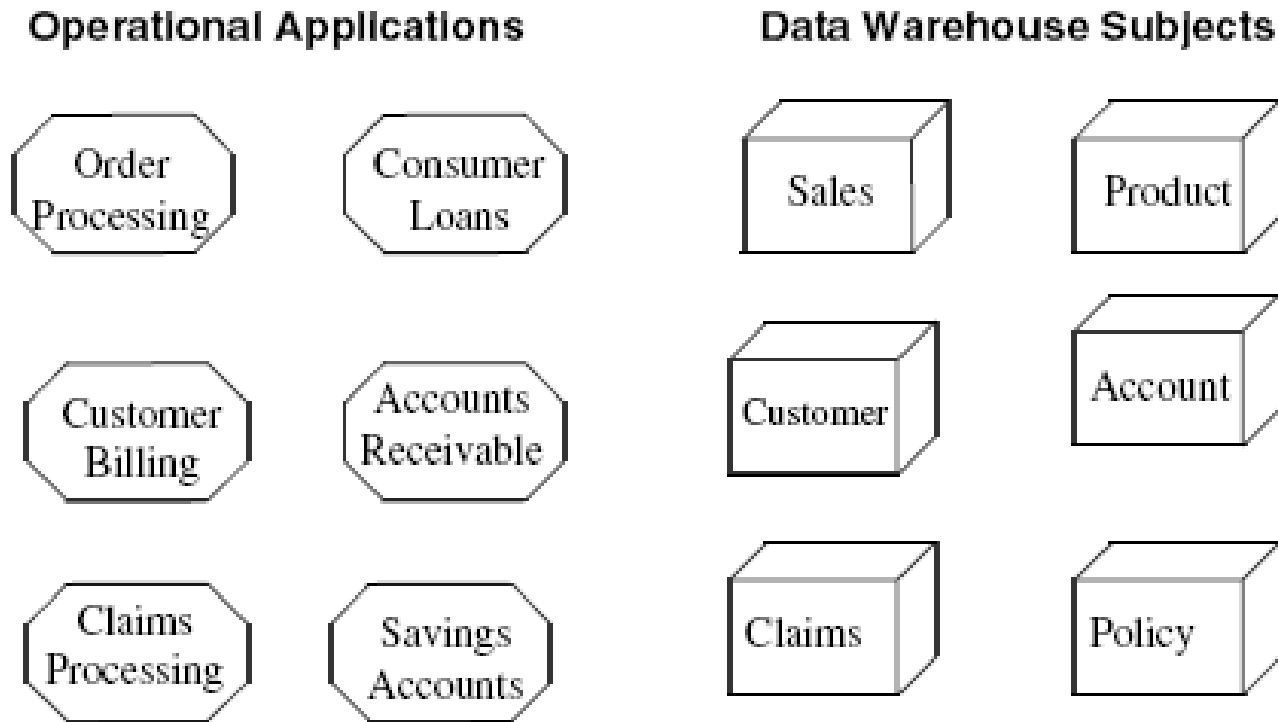


Figure 2-1 The data warehouse is subject oriented.

•Integrated Data

Data inconsistencies are removed; data from diverse operational applications is integrated.

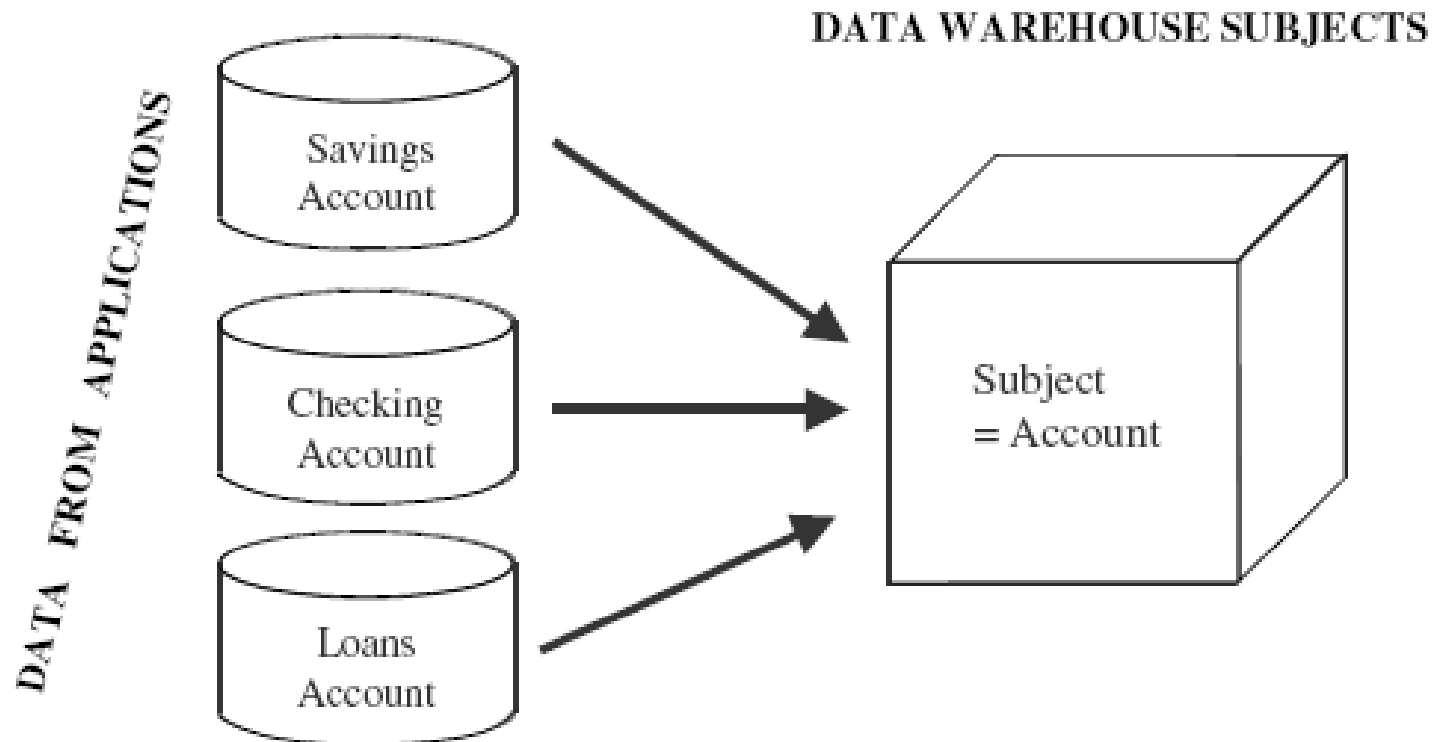


Figure 2-2 The data warehouse is integrated.

•Integrated Data (2)

- Before the data from various disparate sources can be usefully stored in a data warehouse, you have to:
 - remove the inconsistencies;
 - standardize the various data elements;
 - make sure of the meanings of data names in each source application.

•Integrated Data (3)

- Before moving the data into the data warehouse, you have to go through a process of transformation, consolidation, and integration of the source data.
- Here are some of the items that would need standardization:
 - Naming conventions
 - Codes
 - Data attributes
 - Measurements

•Time-Variant Data

- For an operational system, the stored data contains the current values.
- The data in the data warehouse is meant for analysis and decision making.
- A data warehouse, because of the very nature of its purpose, has to contain historical data, not just current values. Data is stored as snapshots over past and current periods. Every data structure in the data warehouse contains the time element.

•Time-Variant Data (2)

- The time-variant nature of the data in a data warehouse
 - Allows for analysis of the past
 - Relates information to the present
 - Enables forecasts for the future

•Nonvolatile Data

Usually the data in the data warehouse is not updated or deleted.

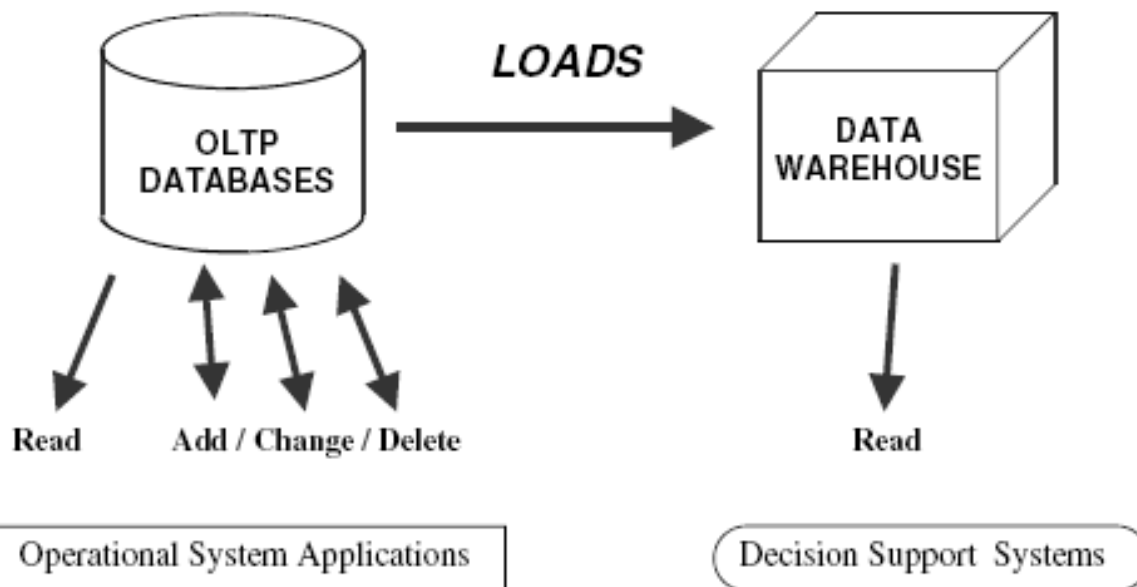


Figure 2-3 The data warehouse is nonvolatile.

•Data Granularity

THREE DATA LEVELS IN A BANKING DATA WAREHOUSE

<u>Daily Detail</u>	<u>Monthly Summary</u>	<u>Quarterly Summary</u>
Account	Account	Account
Activity Date	Month	Month
Amount	Number of transactions	Number of transactions
Deposit/Withdrawal	Withdrawals	Withdrawals
	Deposits	Deposits
	Beginning Balance	Beginning Balance
	Ending Balance	Ending Balance

Data granularity refers to the level of detail. Depending on the requirements, multiple levels of detail may be present. Many data warehouses have at least dual levels of granularity.

Figure 2-4 Data granularity.

• DATA WAREHOUSES AND DATA MARTS

DATA WAREHOUSE	DATA MART
◆ Corporate/Enterprise-wide ◆ Union of all data marts ◆ Data received from staging area ◆ Queries on presentation resource ◆ Structure for corporate view of data ◆ Organized on E-R model	◆ Departmental ◆ A single business process ◆ Star-join (facts & dimensions) ◆ Technology optimal for data access and analysis ◆ Structure to suit the departmental view of data

Figure 2-5 Data warehouse versus data mart.

Top Down Vs Bottom Up Approach

Advantages:

- A truly corporate effort, an enterprise view of data
- Inherently architected—not a union of disparate data marts
- Single, central storage of data about the content
- Centralized rules and contro
- May see quick results if implemented with iterations

DisAdvantages

- Takes longer to build even with an iterative method
- High exposure/risk to failure
- Needs high level of cross-functional skills
- High outlay without proof of concept

Bottom-Up Approach

The advantages of this approach are:

- Faster and easier implementation of manageable pieces
- Favorable return on investment and proof of concept
- Less risk of failure
- Inherently incremental; can schedule important data marts first
- Allows project team to learn and grow

The disadvantages are:

- Each data mart has its own narrow view of data
- Permeates redundant data in every data mart
- Perpetuates inconsistent and irreconcilable data
- Proliferates unmanageable interfaces

- ***OVERVIEW OF THE
COMPONENTS***

Building blocks of DWH

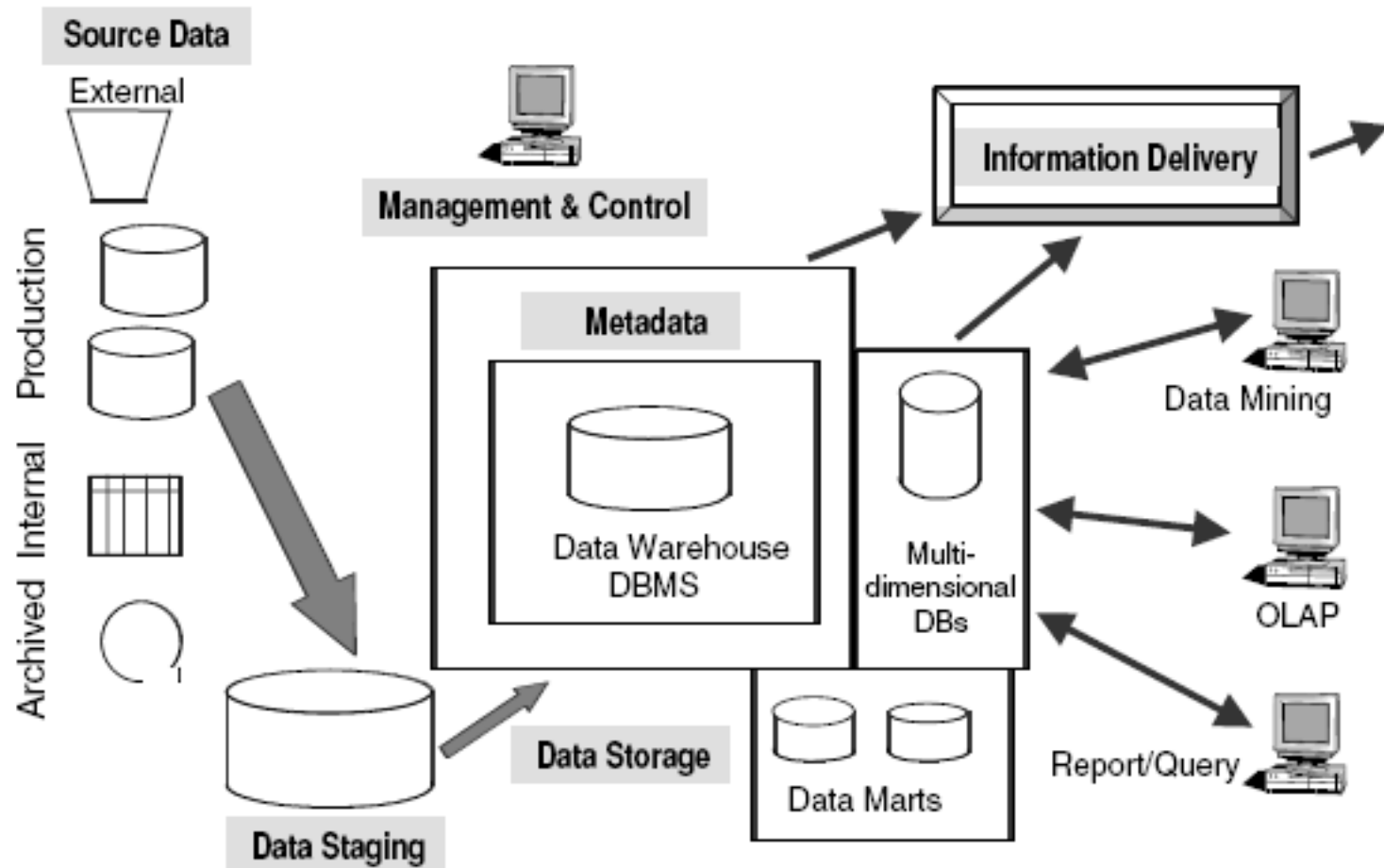


Figure 2-6 Data warehouse: building blocks or components.

Architecture is the proper arrangement of the components.

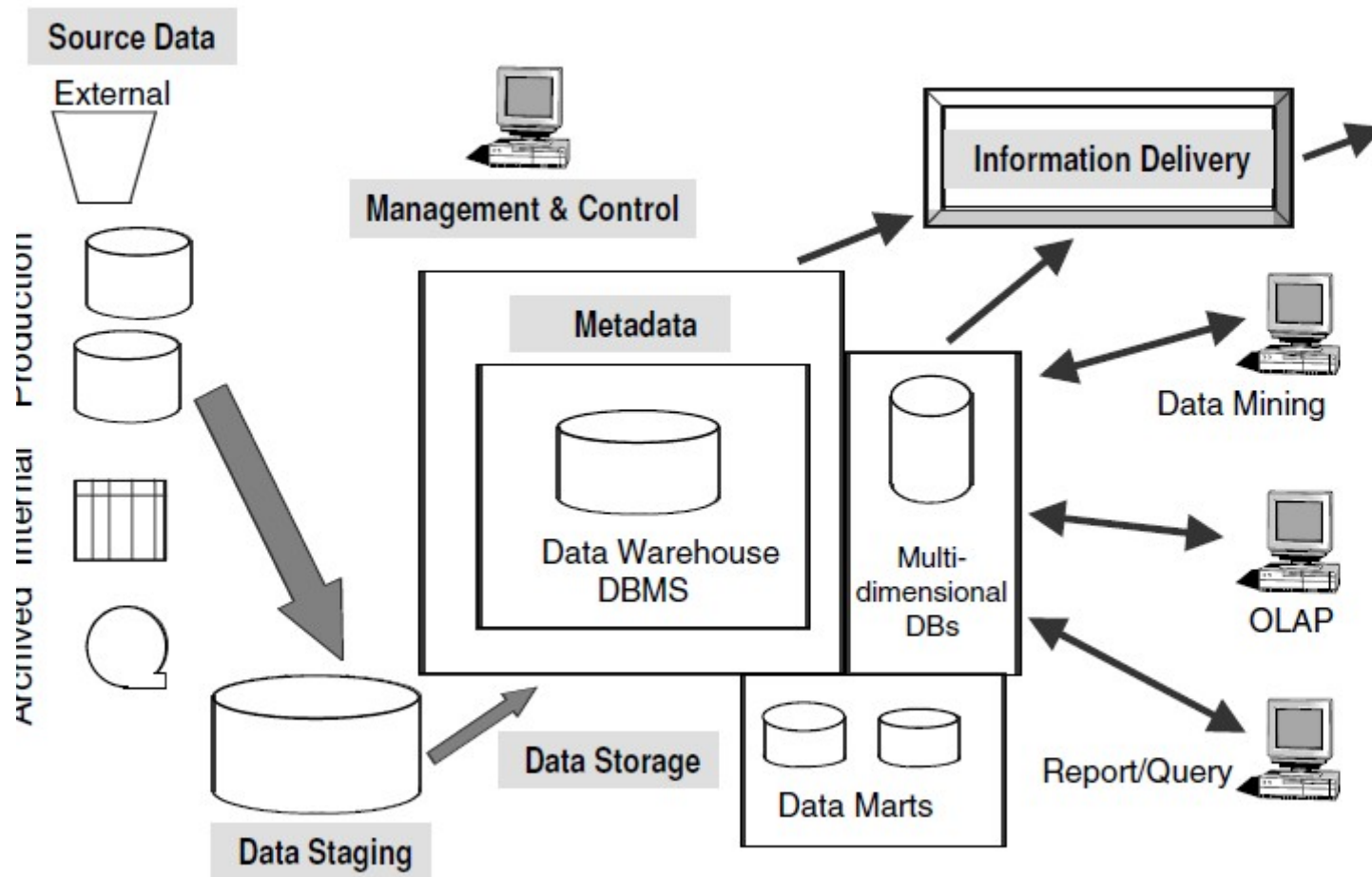


Figure 2-6 Data warehouse: building blocks or components.

•Source data component

- Production systems
- Internal data (spreadsheets)
- Archived data (tapes)
- External data (stocks, interest rates, ...)

•Data Staging Component

- Data Extraction.
- Data Transformation.
- Data Loading.

•Data Movement to the data Warehouse

- ◆ This function is time-consuming
- ◆ Initial load moves very large volumes of data
- ◆ The business conditions determine the refresh cycles

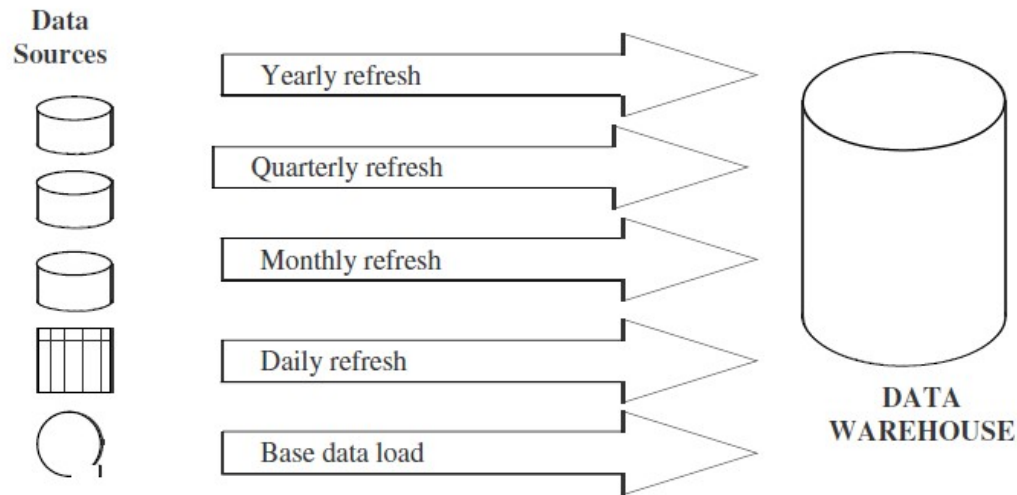


Figure 2-7 Data movements to the data warehouse.

• Information Delivery Component

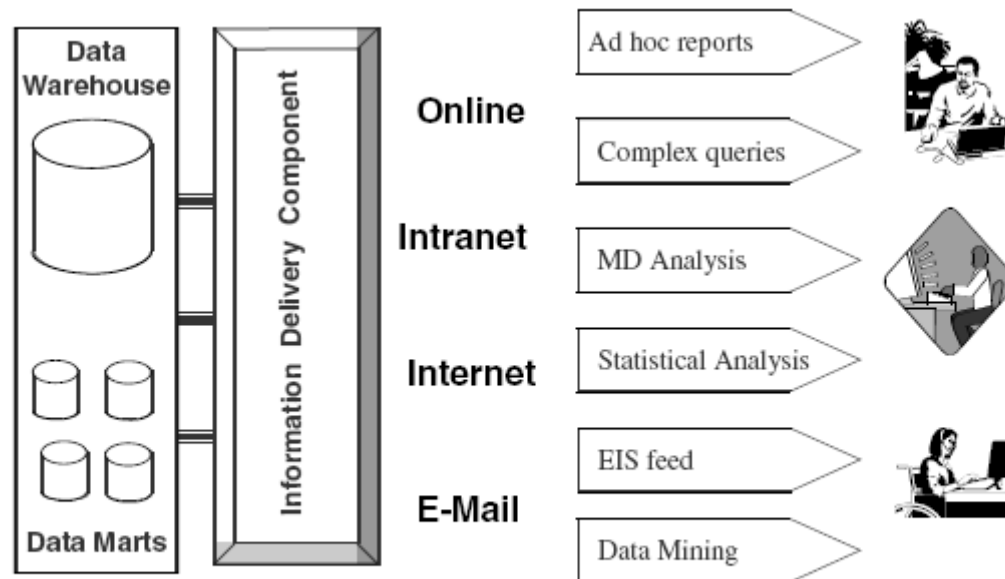


Figure 2-8 Information delivery component.

- **METADATA IN THE DATA WAREHOUSE**

• **WHY METADATA IS IMPORTANT**

• Users to compose and run the query can have several important questions:

- Are there any predefined queries I can look at?
- What are the various elements of data in the warehouse?
- Is there information about unit sales and unit costs by product?
- How can I browse and see what is available?
- From where did they get the data for the warehouse? From which source systems?
- How did they merge the data from the telephone orders system and the mail orders system?
- How old is the data in the warehouse?
- When was the last time fresh data was brought in?
- Are there any summaries by month and product?

- Metadata in a data warehouse contains the answers to questions about the data in the data warehouse.

• Different definitions for metadata

- Data about the data
- Table of contents for the data
- Catalog for the data
- Data warehouse atlas
- Data warehouse roadmap
- Data warehouse directory
- Glue that holds the data warehouse contents together
- Tongs to handle the data
- The nerve center

•Metadata in OLTP

- In operational systems we do not really have any easy and flexible methods for knowing the nature of the contents of the database.
- There is no great need for user-friendly interfaces to the database contents.
- The data dictionary or catalog is meant for IT uses only.

•Metadata in DWH

- Users need sophisticated methods for browsing and examining the contents of the data warehouse.
- Users need to know the meanings of the data items.
- Users have to prevent them from drawing wrong conclusions from their analysis through their ignorance about the exact meanings.
- Without adequate metadata support, users of the larger data warehouses are totally handicapped.

•Types of Metadata

- Metadata in a data warehouse fall into three major categories:
- Operational Metadata
- Extraction and Transformation Metadata
- End-User Metadata

•Operational Metadata

- Data for the data warehouse comes from several operational systems of the enterprise.
- These source systems contain different data structures.
- The data elements selected for the data warehouse have various field lengths and data types.
- In selecting data from the source systems for the data warehouse, you split records, combine parts of records from different source files, and deal with multiple coding schemes and field lengths.
- When you deliver information to the end-users, you must be able to tie that back to the original source data sets.
- Operational metadata contain all of this information about the operational data sources.

• **Extraction and Transformation Metadata**

- Extraction and transformation metadata contain data about the extraction of data from the source systems, namely, the extraction frequencies, extraction methods, and business rules for the data extraction. Also, this category of metadata contains information about all the data transformations that take place in the data staging area.

•End-User Metadata

- The end-user metadata is the navigational map of the data warehouse. It enables the end-users to find information from the data warehouse. The end-user metadata allows the end-users to use their own business terminology and look for information in those ways in which they normally think of the business.

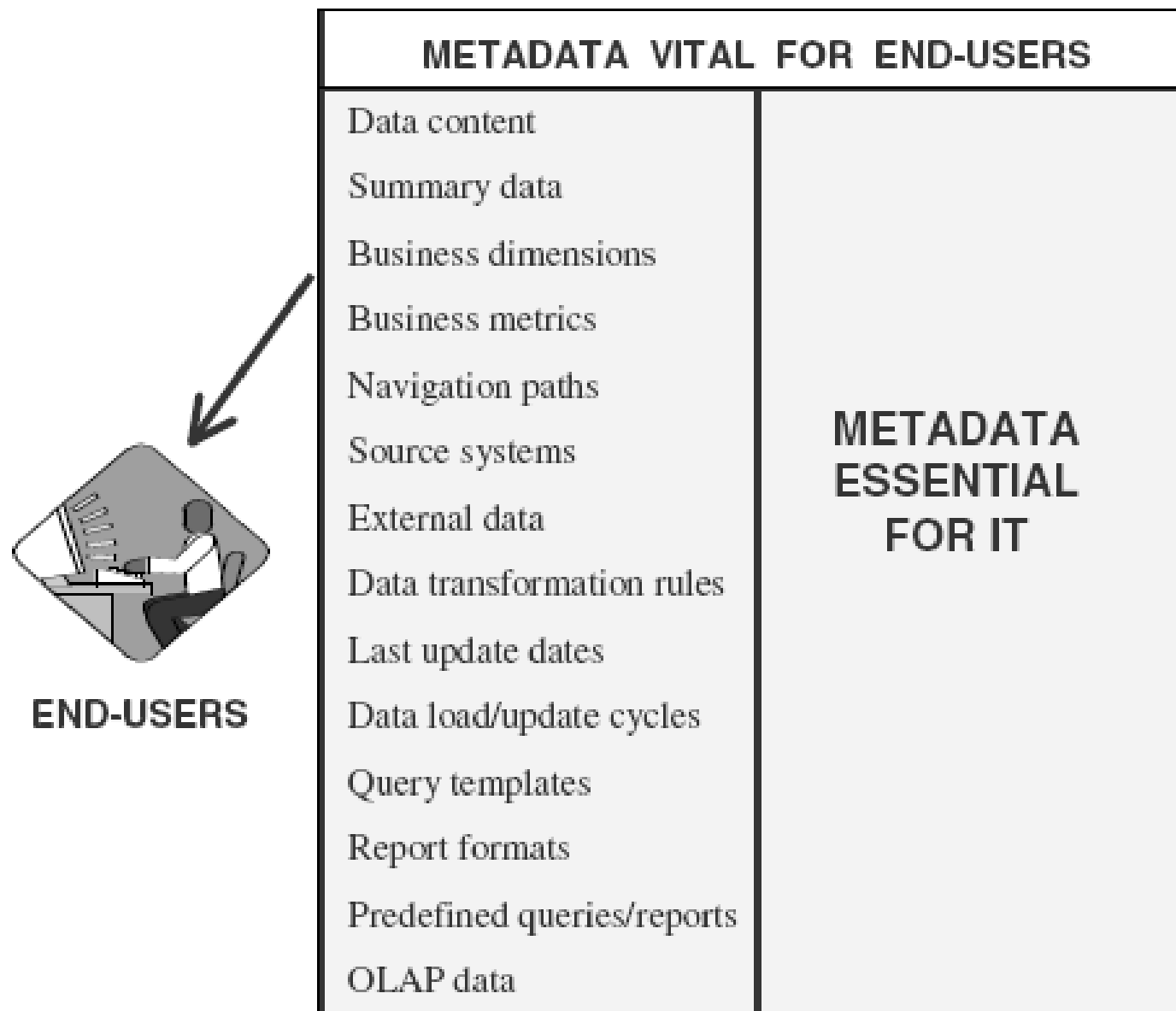


Figure 9-5 Metadata vital for end-users.

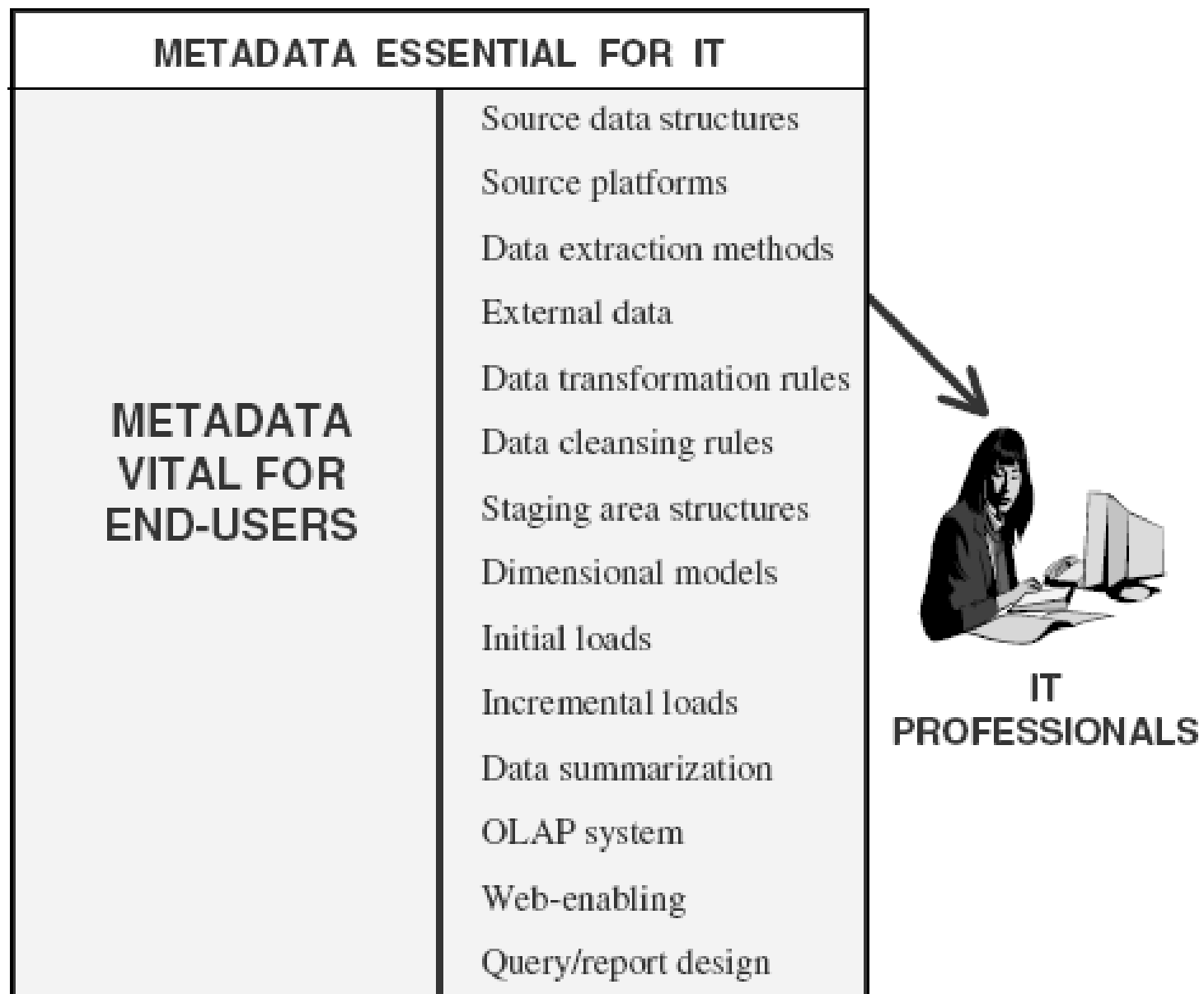


Figure 9-6 Metadata essential for IT.

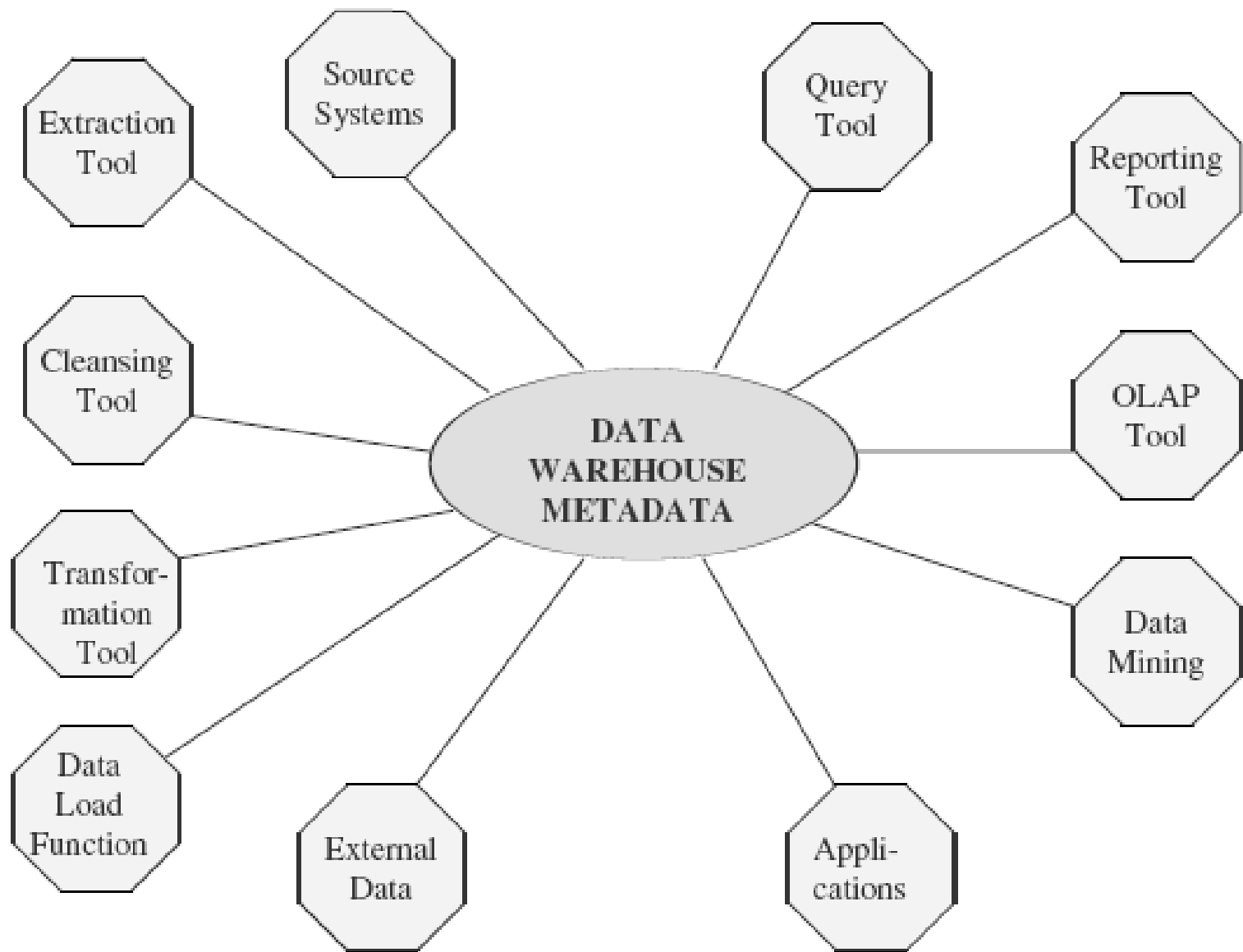


Figure 9-4 Metadata acts as a nerve center.

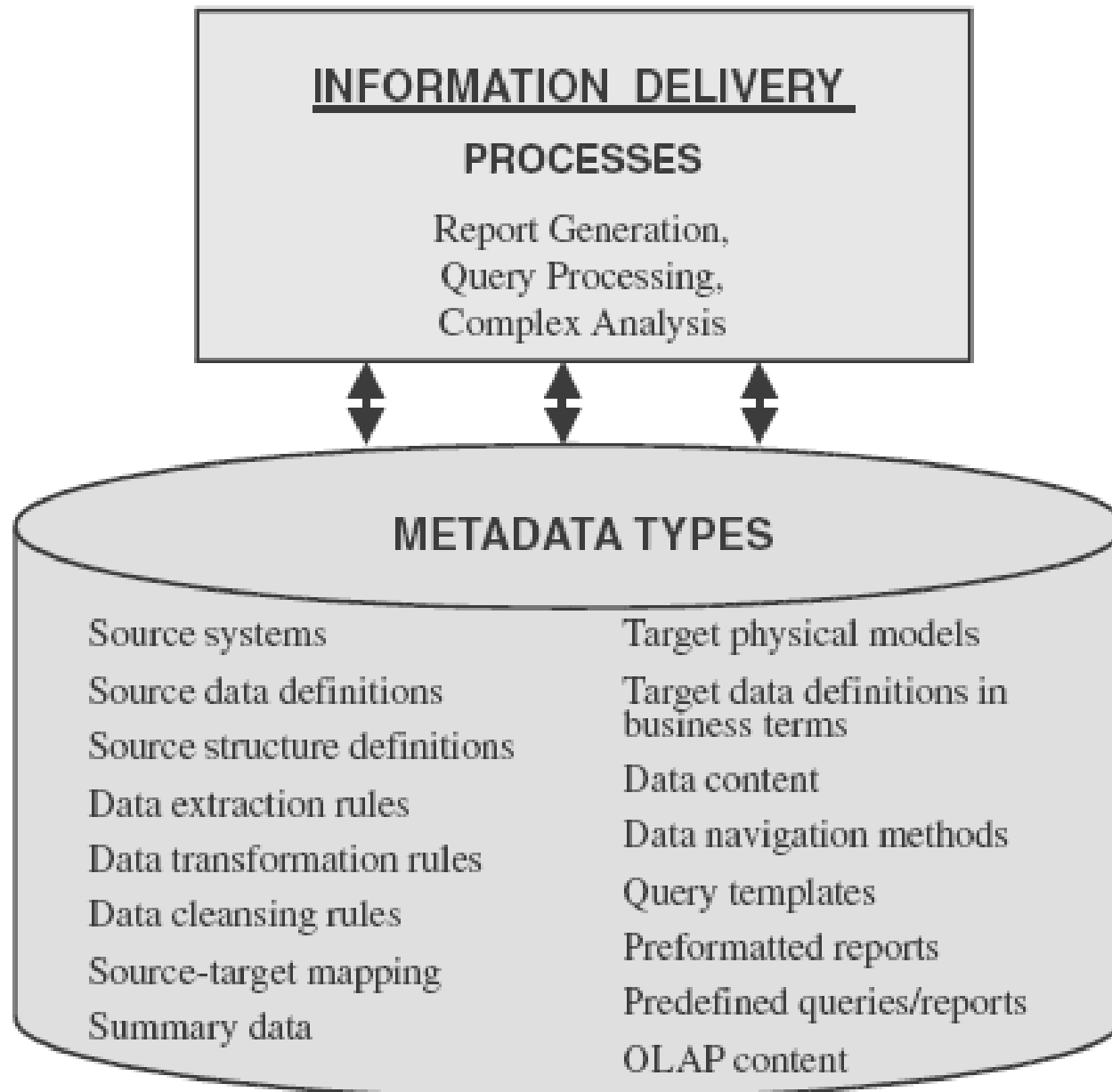


Figure 9-10 Information delivery: metadata types.

- THE ARCHITECTURAL
COMPONENTS

• ARCHITECTURAL FRAMEWORK

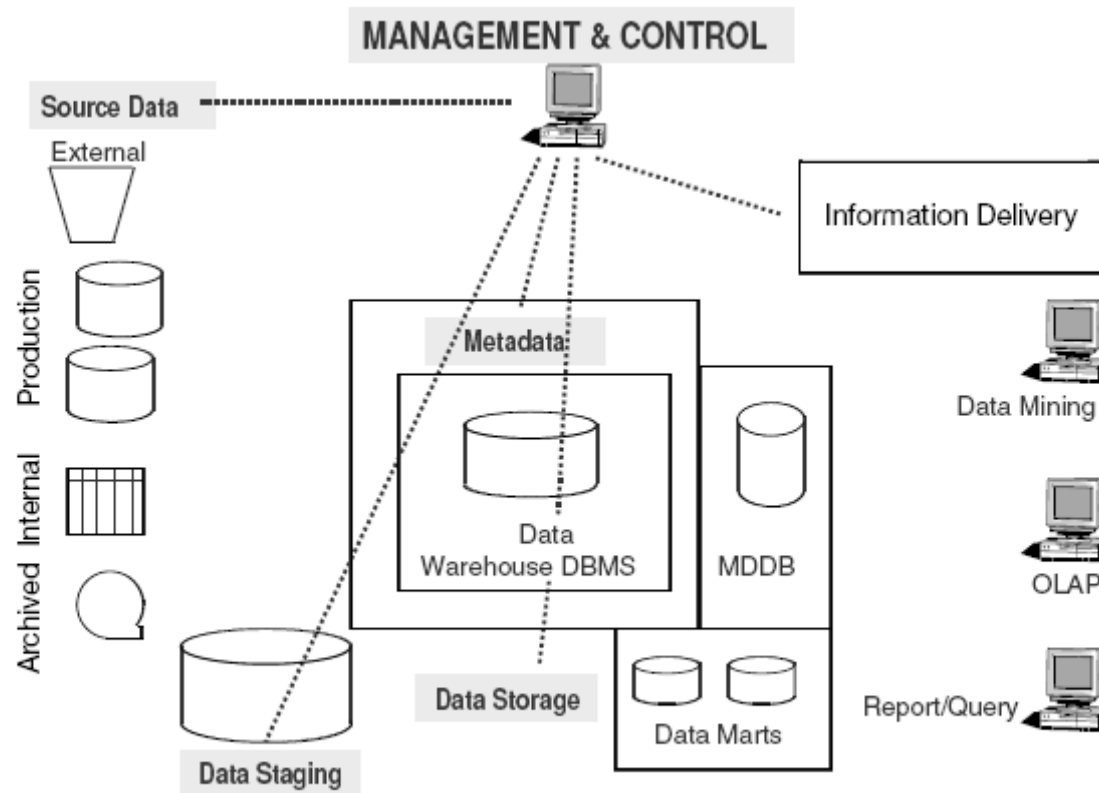


Figure 7-3 The management and control component.

• **TECHNICAL ARCHITECTURE**

- The technical architecture of a data warehouse is the complete set of functions and services provided within its components.
- The technical architecture also includes the procedures and rules that are required to perform the functions and provide the services.
- The technical architecture also encompasses the data stores needed for each component to provide the services.

•Data Acquisition

- Data acquisition covers the entire process of extracting data from the data sources, moving all the extracted data to the staging area, and preparing the data for loading into the data warehouse repository.
- The two major architectural components are *source data* and *data staging*.

•Data Acquisition (2)

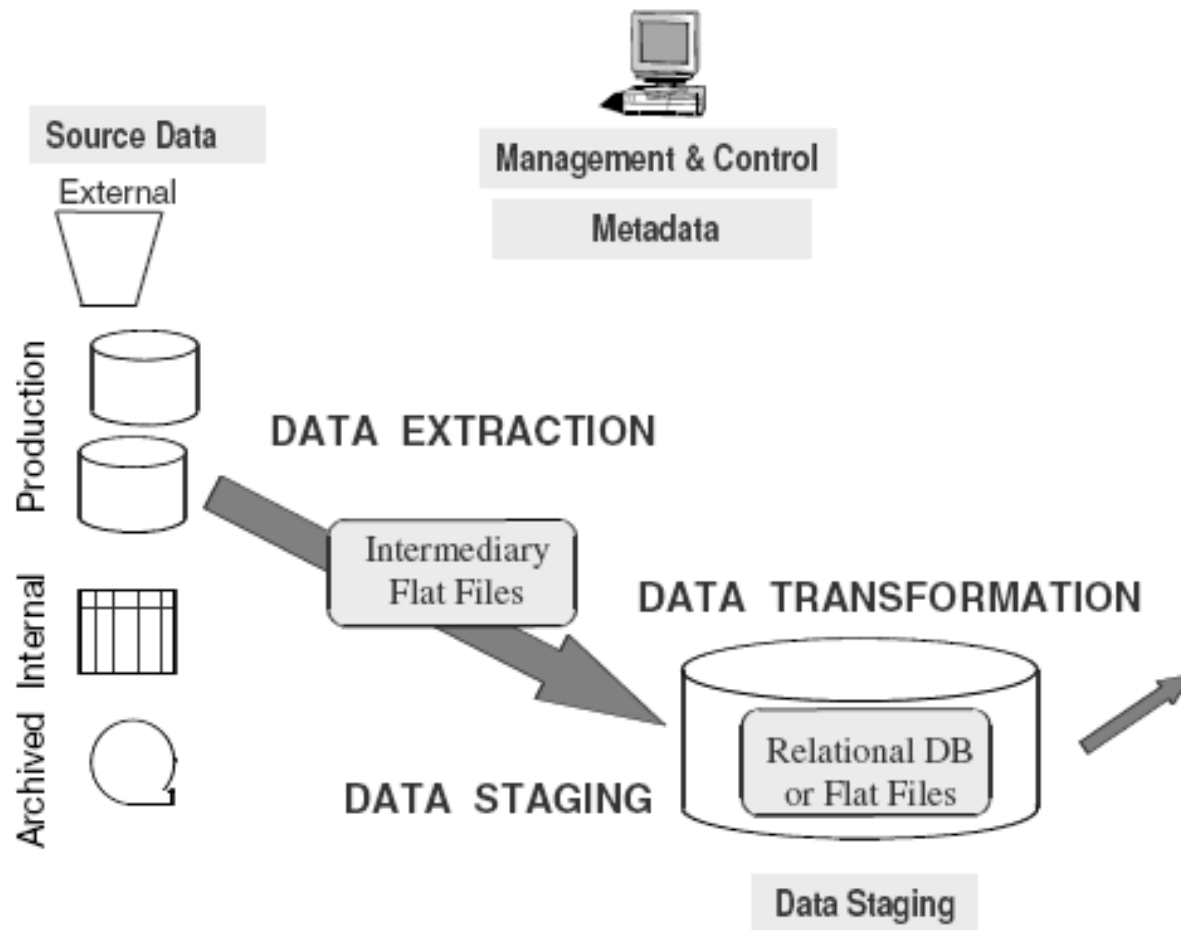


Figure 7-4 Data acquisition: technical architecture.

•List of Functions and Services

- Data Extraction**

- Select data sources and determine the types of filters to be applied to individual sources
- Generate automatic extract files from operational systems using replication and other techniques
- Create intermediary files to store selected data to be merged later
- Transport extracted files from multiple platforms
- Provide automated job control services for creating extract files
- Reformat input from outside sources
- Reformat input from departmental data files, databases, and spreadsheets
- Generate common application code for data extraction
- Resolve inconsistencies for common data elements from multiple sources

•List of Functions and Services (2)

•Data Transformation

- Map input data to data for data warehouse repository
- Clean data, deduplicate, and merge/purge
- Denormalize extracted data structures as required by the dimensional model of the data warehouse
- Convert data types
- Calculate and derive attribute values
- Check for referential integrity
- Aggregate data as needed
- Resolve missing values
- Consolidate and integrate data

•List of Functions and Services (3)

•Data Staging

- Provide backup and recovery for staging area repositories
- Sort and merge files
- Create files as input to make changes to dimension tables
- If data staging storage is a relational database, create and populate database
- Preserve audit trail to relate each data item in the data warehouse to input source
- Resolve and create primary and foreign keys for load tables
- Consolidate datasets and create flat files for loading through DBMS utilities
- If staging area storage is a relational database, extract load files

•Data Storage

- Data storage covers the process of loading the data from the staging area into the data warehouse repository.
- All functions for transforming and integrating the data are completed in the data staging area.
- The prepared data in the data warehouse is like the finished product that is ready to be stacked in an industrial warehouse.

•Data Storage (2)

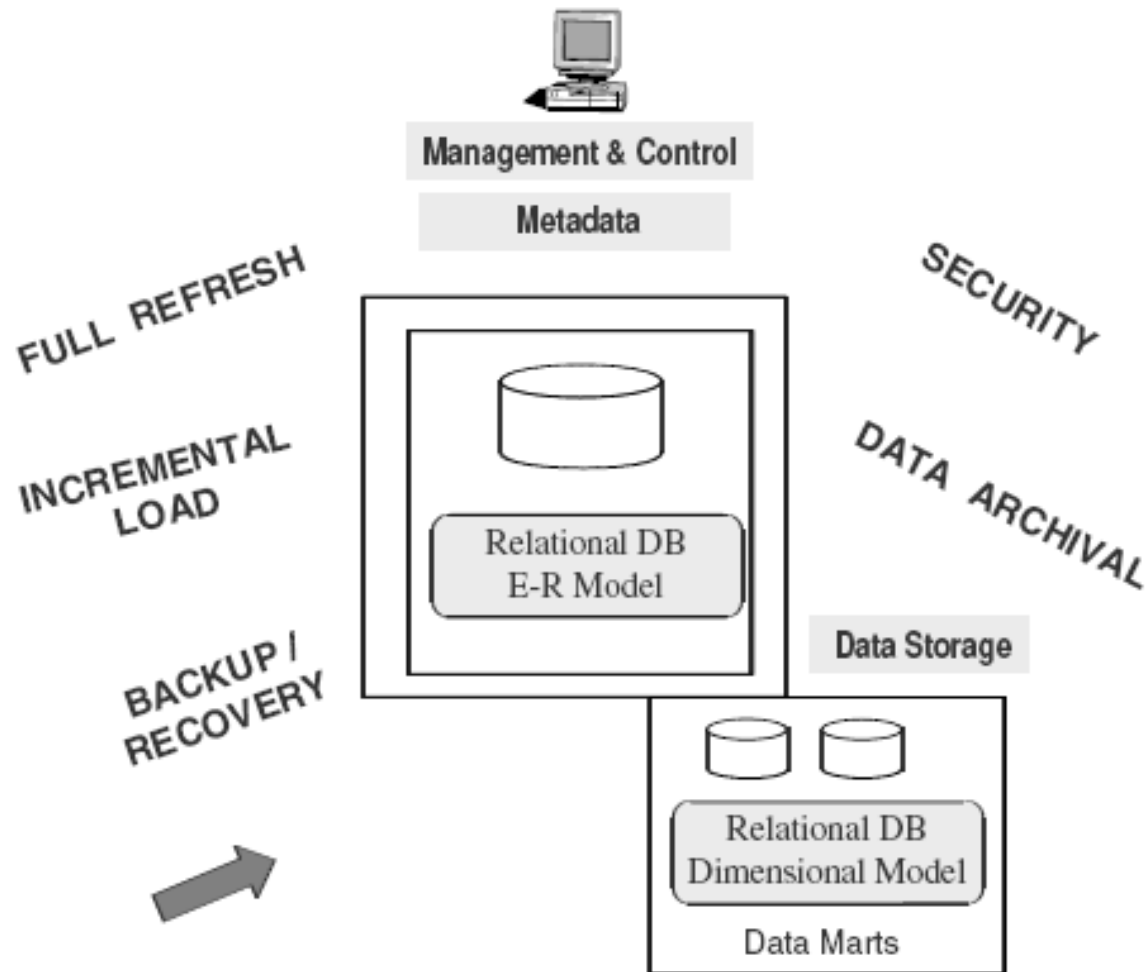


Figure 7-5 Data storage: technical architecture.

•Data Storage (3)

- List of Functions and Services**

- Load data for full refreshes of data warehouse tables
- Perform incremental loads at regular prescribed intervals
- Support loading into multiple tables at the detailed and summarized levels
- Optimize the loading process
- Provide automated job control services for loading the data warehouse
- Provide backup and recovery for the data warehouse database
- Provide security
- Monitor and fine-tune the database
- Periodically archive data from the database according to preset conditions

•Information Delivery

- Information delivery spans a broad spectrum of many different methods of making information available to users.
- For users, the information delivery component is the data warehouse.

•Information Delivery (2)

- The information delivery component makes it easy for the users to access the information either directly from the enterprise-wide data warehouse, from the dependent data marts, or from the set of conformed data marts.
- Most of the information access in a data warehouse is through online queries and interactive analysis sessions.

• Information Delivery (3)

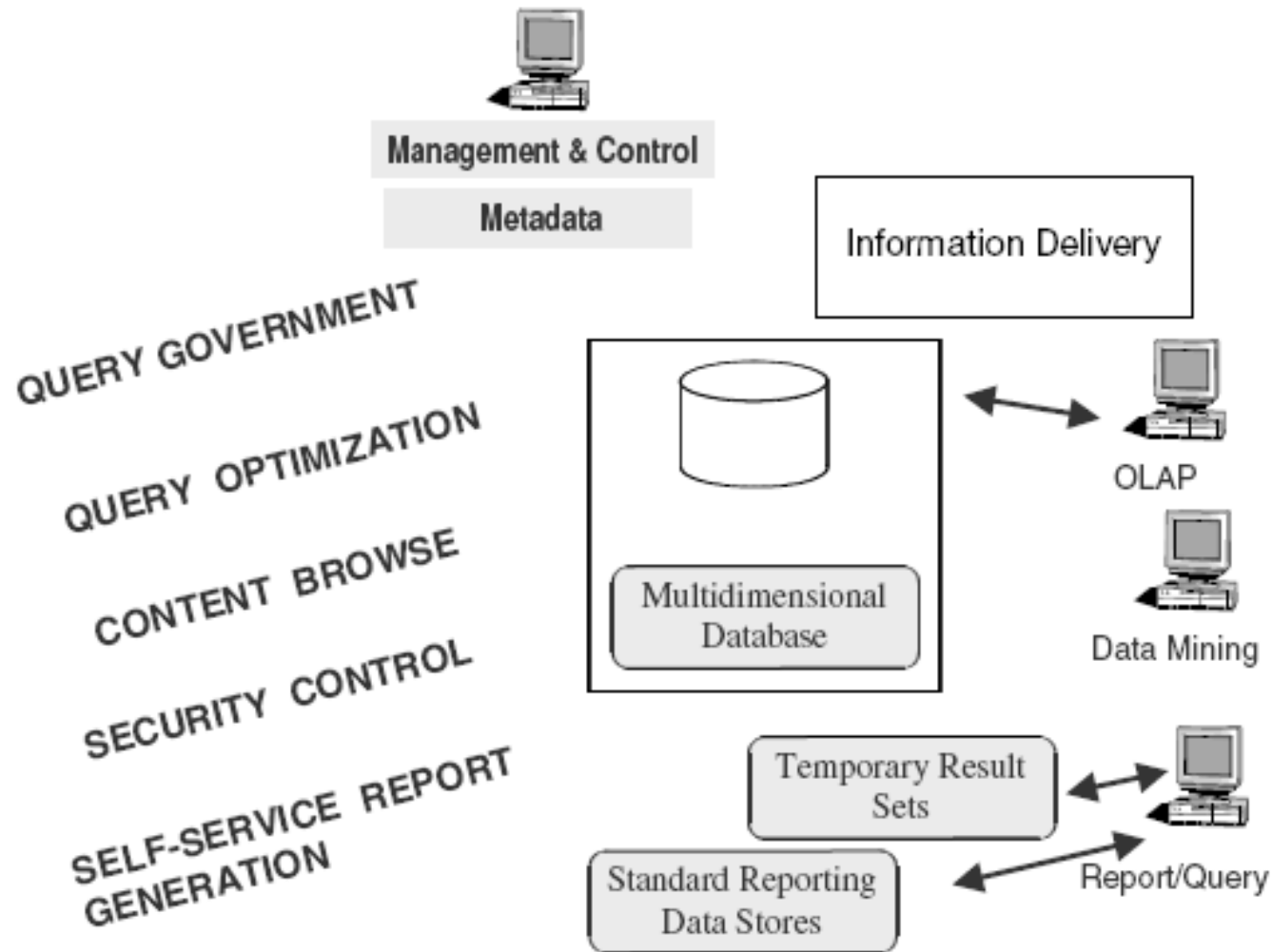


Figure 7-6 Information delivery: technical architecture.

•Information Delivery (4)

- Almost all modern data warehouses provide for online analytical processing (OLAP).
- The primary data warehouse feeds data to proprietary multidimensional databases (MDDBs) where summarized data is kept as multidimensional cubes of information.
- The users perform complex multidimensional analysis using the information cubes in the MDDBs.

• Functions and Services

- Provide security to control information access
- Monitor user access to improve service and for future enhancements
- Allow users to browse data warehouse content
- Simplify access by hiding internal complexities of data storage from users
- Automatically reformat queries for optimal execution
- Enable queries to be aware of aggregate tables for faster results
- Govern queries and control runaway queries
- Provide self-service report generation for users, consisting of a variety of flexible options to create, schedule, and run reports

• Functions and Services (2)

- Store result sets of queries and reports for future use
- Provide multiple levels of data granularity
- Provide event triggers to monitor data loading
- Make provision for the users to perform complex analysis through online analytical processing (OLAP)
- Enable data feeds to downstream, specialized decisions support systems such as EIS and data mining

- Tools

•COLLECTION OF TOOLS

- In a data warehouse environment developers use third-party tools for different phases of the development
 - code-generators for preparing in-house software for data extraction
 - accessing information through third-party query tools
 - creating reports with report writers

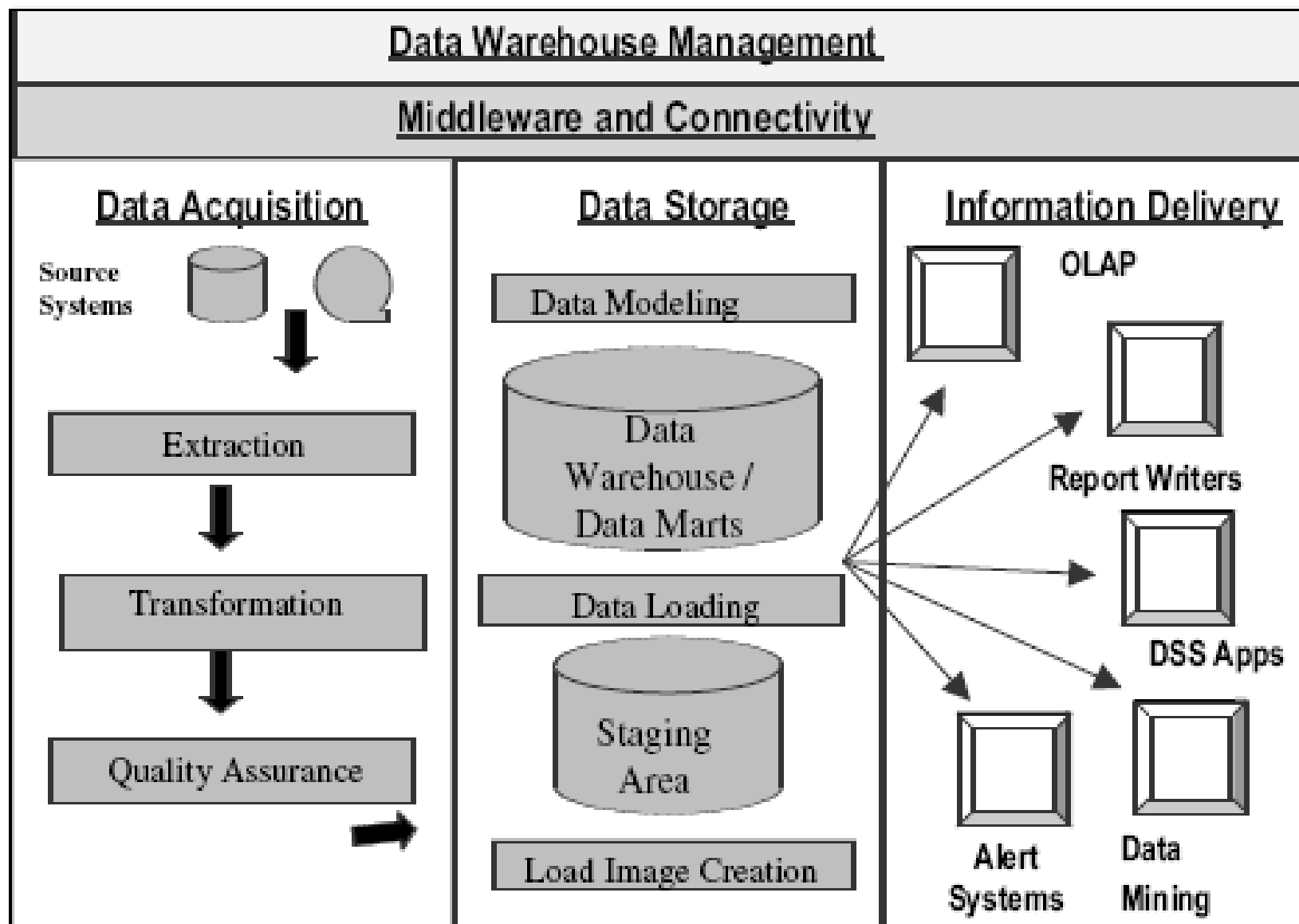


Figure 8-16 Tools for your data warehouse.

- **Basic purposes and features of the type of tool**

•Data Modeling

- Enable developers to create and maintain data models for the source systems and the data warehouse target databases. If necessary, data models may be created for the staging area.
- Provide forward engineering capabilities to generate the database schema.
- Provide reverse engineering capabilities to generate the data model from the data dictionary entries of existing source databases.
- Provide dimensional modeling capabilities to data designers for creating STAR schemas

•Data Extraction

- Two primary extraction methods are available: bulk extraction for full refreshes and change-based replication for incremental loads.
- Tool choices depend on the following factors: source system platforms and databases, and available built-in extraction and duplication facilities in the source systems.

•Data Transformation

- Transform extracted data into appropriate formats and data structures.
- Provide default values as specified.
- Major features include field splitting, consolidation, standardization, and deduplication.

•Data Loading

- Load transformed and consolidated data in the form of load images into the data warehouse repository.
- Some loaders generate primary keys for the tables being loaded.
- For load images available on the same RDBMS engine as the data warehouse, precoded procedures stored on the database itself may be used for loading.

•Data Quality

- Assist in locating and correcting data errors.
- May be used on the data in the staging area or on the source systems directly.
- Help resolve data inconsistencies in load images.

•Queries and Reports

- Allow users to produce canned, graphic-intensive, sophisticated reports.
- Help users to formulate and run queries.
- Two main classifications are report writers, report servers.

•Online Analytical Processing (OLAP)

- Allow users to run complex dimensional queries.
- Enable users to generate canned queries.
- Two categories of online analytical processing are multidimensional online analytical processing (MOLAP) and relational online analytical processing (ROLAP).
 - MOLAP works with proprietary multidimensional databases that receive data feeds from the main data warehouse. ROLAP provides online analytical processing capabilities from the relational database of the data warehouse itself.